Evaluation of the Unit 11 project vs. the Unit 6 project

1. Introduction

The two projects demonstrate my progress and knowledge of machine learning throughout the module. The team project submitted in Unit 6 focuses on developing regression and clustering models to identify price patterns and listing segments, while the project submitted in Unit 11 compares a traditional machine learning SVD model with a deep learning model to build a movie recommender system, incorporating explainable AI, bias analysis, and deployment considerations.

2. Comparison between Project Unit 6 and Project Unit 11

Several differences and developments can be identified when comparing the Unit 6 project with the Unit 11 project.

In the first project, the aim was to develop regression and clustering models to identify price patterns and listing segments. It was a group project, meaning that the tasks were divided among six people, and consequently, there was the possibility of discussing uncertainties and decisions together.

The final project was an individual project in which we had to compare a traditional machine learning model with a deep learning model. I chose to create two recommender systems that recommend movies based on users' ratings. In this case, being alone and trying to develop two machine learning models was more difficult because I had to make all decisions independently, and I had no one to discuss the project with.

In the Unit 6 project, my main responsibilities were exploratory data analysis (EDA) and data cleaning. My role was to provide the group with an initial analysis of the data, and a high-quality dataset would be essential to the subsequent stages of the project.

After completing my task, I continued to participate in weekly meetings to discuss the challenges encountered during the project and support the team by reviewing the results. Although the technical challenges were manageable, the main difficulty was group coordination.

The final project compared two different models: one traditional machine learning model and one deep learning model.

For the traditional machine learning model, an SVD-based matrix factorization was selected, following Koren et al. (2009), who stated that it was suitable for datasets such as MovieLens because it projects users and items into a shared hidden representation space, thereby capturing user preferences and characteristics and providing accurate recommendations. For the deep learning model, a Neural Collaborative Filtering model was selected. According to He et al. (2017), it combines user and item embeddings through a Multi-Layer Perceptron and captures richer latent relationships that may improve the recommendation performance.

Overall, the final project in Unit 11 represents a significant development from the Unit 6 initial project in terms of complexity, methodology, and project management.

Starting with project management, in the initial project, responsibilities and decisions were shared among the group members, whereas in the final project, all decisions were made individually.

In addition, the Unit 11 assignment involves higher technical complexity than the Unit 6 project. The final project required a comparison between two machine learning models, which needed to be implemented using a real dataset. It also included RMSE and MAE evaluation, explainability, bias analysis, and deployment, while the initial proposal focused on the implementation of linear regression and clustering models. From this comparison, it can be concluded that the Unit 11 assignment is more comprehensive and aligned with a real-world machine learning workflow. Furthermore, it required a critical comparison of models using evaluation criteria, while the initial project was closer to a data analysis and model development project.

Finally, both projects achieved their goals, as the Unit 6 project resulted in a report in which the group presented the development and analysis of linear regression and clustering models. In contrast, the Unit 11 presentation included the implementation of two machine learning models and a comparison between them. It is possible to conclude that the Unit 11 presentation is more complete because it considers more stages of the machine learning development process.

3. Conclusion

In conclusion, both projects meet all the assignment requirements. When comparing them, it is possible to state that the Unit 11 project is an evolution of the Unit 6 project. It requires students to have more autonomy, more technical skills, a more comprehensive workflow, and a comparison between models using evaluation criteria, bringing it closer to a complete machine learning development process used in real-world applications.

Reference list

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